Intrinsic Motivation vs. Entropy Regularization: A Controlled Comparison of On-Policy and Off-Policy RL Under Sparse Rewards in Classic Control

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1. Motivation

Sparse rewards remain one of the central challenges in deep reinforcement learning: an agent may interact with an environment for thousands of steps before receiving any learning signal, making policy optimization slow and unstable. Two broad families of solutions have emerged — *entropy regularization*, which encourages exploratory stochastic policies by adding a policy-entropy bonus to the objective, and *Random Network Distillation (RND)*, which augments the extrinsic reward with a novelty bonus derived from prediction error. Both are widely used in practice, yet they are rarely evaluated under matched compute conditions and controlled reward-density settings, making it difficult to reason about which mechanism provides greater benefit and when.

A successful project would enable practitioners to make principled algorithmic choices when deploying on-policy (PPO) or off-policy (DQN) agents in sparse-reward environments, targeting a concrete metric: characterising the regime where RND reduces episodes-to-threshold by ≥20% relative to entropy-only baselines.

2. Gap in the Literature

Prior work. Mnih et al. (2015) established DQN as a strong off-policy baseline on discrete-action tasks. Schulman et al. (2017) introduced PPO with clipped surrogate objectives and demonstrated competitive sample efficiency. Burda et al. (2019) proposed RND as a scalable intrinsic reward for sparse-reward exploration. These works differ in evaluation environments, reward structures, and compute budgets, making direct comparison difficult.

What is missing. No controlled study isolates the contribution of entropy regularization versus RND-based intrinsic motivation across both on-policy and off-policy algorithms under matched compute and identical reward-density conditions. Prior comparisons either fix the algorithm family or do not ablate the exploration mechanism independently.

Our angle. *No controlled study compares entropy regularization and RND intrinsic motivation across PPO and DQN under matched wall-clock compute on Gymnax classic-control environments with both sparse and dense reward variants.*

3. Research Questions

Primary: Under sparse rewards and matched compute, does RND intrinsic motivation improve sample efficiency (episodes to threshold return) more for on-policy PPO than for off-policy DQN, or is the gap closed by entropy regularization alone?

Secondary: Does the relative advantage of RND over entropy regularization diminish as reward density increases from sparse to dense, and does this effect differ between PPO and DQN?

4. Experimental Methodology

Environment & Data. We use Gymnax (JAX-native) CartPole-v1 and MountainCar-v0. Both environments support fast, parallelisable rollouts and are well-understood benchmarks. Sparse-reward variants are constructed by replacing the dense reward with a binary signal (success/failure at episode end) to control reward density as an independent variable.

Algorithms & Conditions (6 total).

- **PPO** — on-policy baseline
- **PPO + Entropy Regularization** ($\alpha \in \{0.01, 0.05\}$)
- **PPO + RND** — on-policy with intrinsic reward
- **DQN** — off-policy baseline
- **DQN + Entropy Regularization** (ε-schedule + entropy term)

- **DQN + RND** — off-policy with intrinsic reward

All conditions share identical MLP architectures (2 hidden layers, 64 units, ReLU) and are run for the same wall-clock budget (2 hours per condition on a personal GPU).

Controls & Baselines. A random-action agent and a hand-coded heuristic agent serve as lower bounds. Vanilla PPO and vanilla DQN are the primary baselines. All methods use identical evaluation protocols (greedy rollouts every 50 training episodes).

Hyperparameter Search. Learning rates ; discount factor (γ); entropy coefficient (α); RND reward scale (β) . Selection by validation return on a held-out 20% of seeds; fixed budget of 30 configurations.

Seeds & Statistics. 10 independent seeds per condition. Results reported as mean ± standard error with individual seed traces included. Welch's t-test used for pairwise significance.

Metrics. Average return vs. episodes; episodes-to-threshold (90% of max return); policy entropy over training; RND prediction error over training.

Compute & Reproducibility. Personal GPU (NVIDIA). JAX ; pinned versions in environment.yml and requirements.txt. Public GitHub repo with README, deterministic run scripts, and pre-trained checkpoints. All experiments reproducible from a single shell command.

GitHub Link: https://github.com/mohammad1774/elg5214_group_8_project

Risk & Mitigation. Risk: DQN training instability in sparse-reward MountainCar. Mitigation: gradient clipping (norm 10), prioritised experience replay, and fallback to CartPole-only or other simpler game if MountainCar results are unreliable.

5. Expected Contributions

- A reproducible, matched-compute comparison clarifying when RND outperforms entropy regularization for both PPO and DQN under sparse and dense rewards.
- Clean JAX implementations of PPO, DQN, RND, and entropy regularization, reusable as baselines by the broader community.
- An ablation over reward density revealing the crossover point at which intrinsic motivation ceases to provide a statistically significant advantage.

6. Work Distribution

Students A will lead the literature review and finalise the research questions. **Student A & C** will implement and validate the PPO variants (baseline, entropy, RND). **Student B & D** will implement and validate the DQN variants (baseline, entropy, RND). **All members** will contribute to experiment design, hyperparameter sweeps, statistical analysis, and report writing. Code review will be cross-assigned (A reviews D's code, B reviews C's code and vis a vis) to ensure correctness.

References

[1] Mnih, V. et al. (2015). Human-level control through deep reinforcement learning. Nature, 518, 529–533.

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[4] Lax, M. et al. (2023). Gymnax: A JAX-based Reinforcement Learning Environment Library. arXiv:2311.16943.